4.2.1 Uses of Primary Cell

Primary cells are used in electronic products ranging from watches, smoke alarms, cardiac pacemaker torches, hearing aids, transister radios, etc.

4.2.2 Series Cell Connection

Cells are connected in series by connecting the positive terminal of one cell to the negative terminal of the next cell. (See the connection diagram in fig. 4.3)

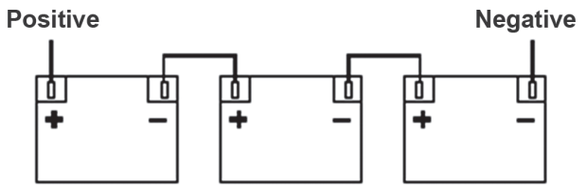


Fig 4.3 *Batteries Series Connection*

Identical cells are connected in series to obtain a higher voltage is available as a single cell. With this connection of cells, the output voltage is equal to the sum of the voltages in the cells.However, the ampere hour (A-h) rating remains equal to that of a single cell.

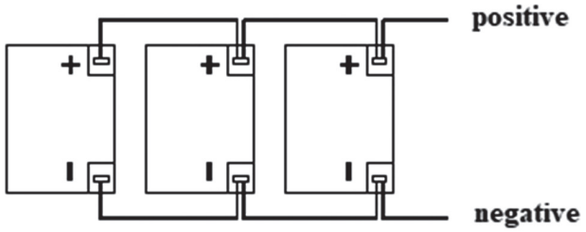


Fig 4.4 *Batteries Parallel Connection*

4.2.3 Parallel Connection

Cells are connected in parallel by connecting all the positive terminals together and all the negative terminals together as shown in the figure 4.4.

Identical cells are connected in parallel to obtain a higher output current or ampere-hour rating. With this connection of cells, the output ampere-hour rating is equal to the sum of the ampere-hour rating of all the cells. However, the output voltage remains the same as that of a single cell.

When connecting groups of cells or batteries in parallel, each group must be in the same voltage level paralleling two batteries of unequal voltage levels set up a difference of potential energy between the

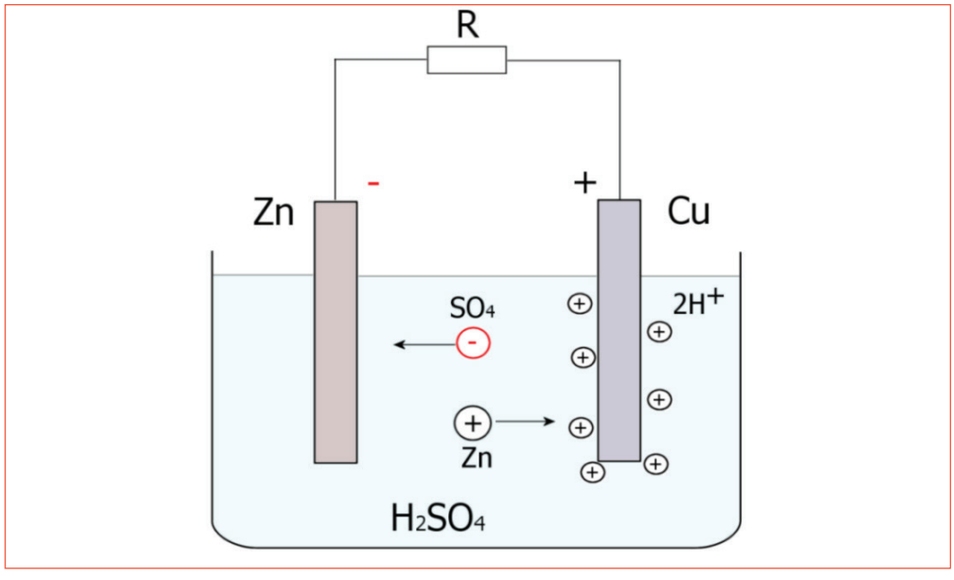


Fig 4.5 *Voltaic Cell*

EMA_English_Chp04.indd 57 11/9/2020 3:01:02 PM